

Muhammad Ibrahim

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TECHNICAL SKILLS

- **AI & LLM Systems:** GPT-4, Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Function Calling, Conversation Memory, OpenRouter Frameworks
- **Backend Development:** FastAPI, Node.js, Redis, WebSockets, JWT, Docker Containerization
- **Frontend Engineering:** React.js, Next.js, TypeScript, Responsive Web Design
- **Languages:** Python, JavaScript, SQL (PostgreSQL/MySQL)
- **Data Systems & Tools:** ETL Pipelines, Data Validation, Git Version Control, Tableau, SSIS

EXPERIENCE

IBEX

Software Engineer

Lahore, Pakistan

Nov 2024 – Present

- Engineered a production-ready AI Agent Builder Platform with robust voice and chat interfaces, enabling internal teams and prompt engineers to rapidly build, test, and demo intelligent agents.
- Designed and built conversation history tracking, complex function-calling orchestrations, and Redis-based session management to handle context-aware, low-latency agent interactions.
- Developed scalable full-stack features using React, Node.js, and FastAPI to provide highly configurable agent deployment workflows.
- Standardized enterprise AI deployment across teams by successfully onboarding legacy internal tools, including service desk bots, directly onto the new platform.
- Collaborated with core AI teams on RAG-based architectures to integrate enterprise-wide, domain-specific knowledge bases into agent workflows.

PROJECTS

AI Agent Builder Platform (AMP)

Platform Link

Tech Stack: Python, FastAPI, React.js, OpenAI API, Redis, WebSockets, Docker

- Architected a comprehensive platform tailored for defining and deploying configurable AI agents featuring native voice and chat compatibility.
- Programmed custom conversational workflows using advanced function calling and custom RAG knowledge pipelines for domain-aware responses.
- Containerized applications utilizing Docker to ensure seamless, repeatable infrastructure environments across development and production environments.

MeetBot – AI-Driven Meeting Assistant

GitHub · Live Demo

Tech Stack: Next.js, TypeScript, FastAPI, RAG, OpenRouter, Notion API

- Developed an end-to-end meeting assistant application that transforms raw meeting transcripts into a structured, searchable knowledge base.
- Built a responsive Next.js and TypeScript frontend coupled with a high-performance FastAPI backend to handle high-throughput transcript ingestion.
- Implemented speaker-aware chunking strategies and vectorized retrieval (RAG) with LLM-driven summarization using the OpenRouter API.
- Integrated asynchronous background processing jobs for data extraction and engineered a Notion API connection to automatically export summaries and streamline post-meeting action items.

EDUCATION

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelor of Science in Computer Science (CGPA: 3.42 / 4.0)

Oct 2020 – Jun 2024

- **Final Year Project:** Suspicious Behavior Detection using CCTV – Awarded **3rd Position** at the Department Industrial Open House for excellence in real-time computer vision modeling.